VCBench Founder Success Prediction: Comparing Feature Generation Approaches

Summary

We compare three approaches to feature engineering for predicting founder success on VCBench:

1. **Human Engineering**: 28 hand-crafted features from structured profile data
2. **LLM Engineering**: 17 features generated by an LLM reasoning pipeline
3. **Policy Induction**: 16 features derived from RL-discovered evaluation policies

While no single approach significantly outperforms the others in isolation, combinations involving Policy Induction significantly beat both baselines. The best model (LLM Engineering + Policy Induction) achieves test F0.5 = 0.334, significantly outperforming Human Engineering ($p = 0.007$) and LLM Engineering ($p = 0.028$). Policy Induction captures unique predictive signal with low feature overlap to other approaches, explaining why it combines so effectively.

---

1. Setup

Dataset

VCBench contains ~9,000 anonymised founder profiles sourced from LinkedIn and Crunchbase. Each founder's company raised between $100K and $4M in funding. A founder is labelled "successful" if their company achieved either $500M+ total funding or a $500M+ exit/IPO (approximately 9% of the dataset).

The data is split into a 4,500-founder training pool and a 4,500-founder held-out test set. Experiments involving LLM-Engineered features use a slightly smaller 4,400-founder training pool (100 founders reserved as LLM seed examples), but all experiments are evaluated on the same test set.

Three Feature Generation Approaches

1. **Human Engineering (HQ)**: 28 hand-crafted features from structured profile data (seniority levels, education flags, exit counts, career trajectory metrics). Model: XGBoost with tuned hyperparameters.

2. **LLM Engineering**: 17 features generated by an LLM reasoning pipeline that examines founder profiles and produces binary/continuous features (e.g., `has_high_qs_education`, `has_public_experience`). Model: L2-regularised logistic regression with nested CV for regularisation strength.

3. **Policy Induction**: 16 policies discovered through RL-guided policy search. Each policy is a natural-language rule (e.g., "founders with long histories founding small businesses are less likely to succeed at scale"). Gemini evaluates each founder against each policy, producing a probability score P(success | policy, founder). Model: L2-regularised logistic regression with nested CV.

Evaluation

- **Primary metric**: F0.5 (precision-weighted, reflecting the VC screening use case where precision matters more than recall)

- **Cross-validation**: 5-fold stratified CV on the training pool, with nested inner 3-fold CV for hyperparameter selection (see Appendix B for regularisation details)
- **Test set**: Held-out 4,500 founders, used only for final evaluation and statistical significance testing (see Appendix D for full methodology)

---

2. Main Results

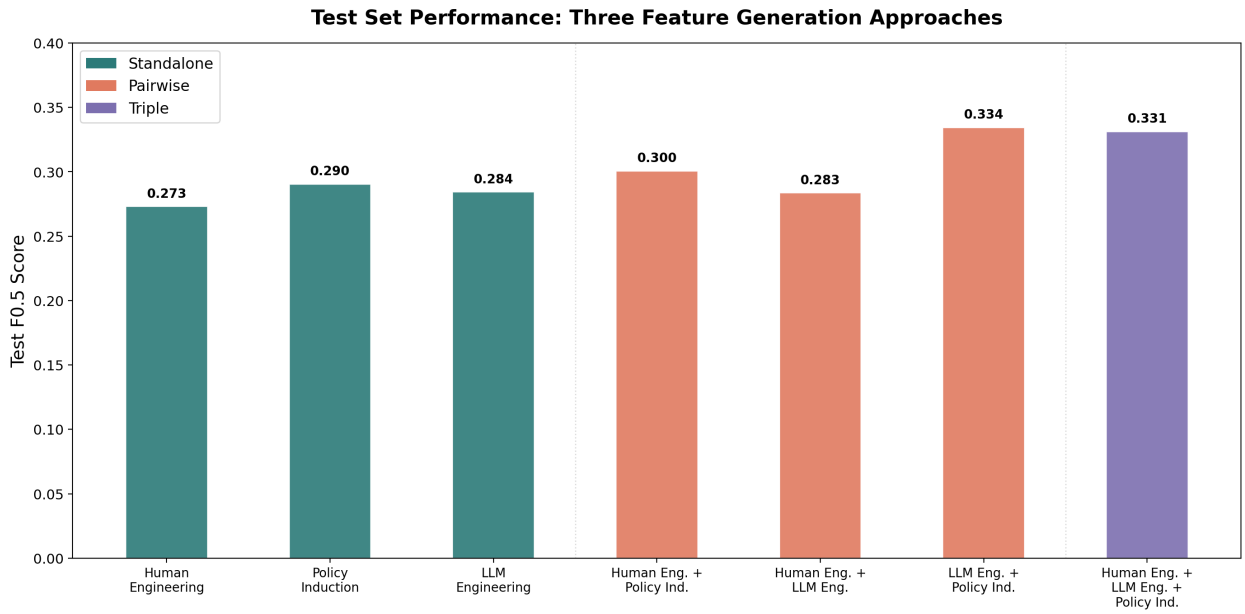

Figure 1: Test F0.5 scores for all 7 experiments.

Exp	Features	# Feat	CV F0.5	Test F0.5	Test Prec	Test Recall
1	Human Engineering	28	0.210	0.273	0.277	0.257
2	Policy Induction	16	0.309	0.290	0.396	0.141
3	LLM Engineering	17	0.266	0.284	0.301	0.232
4	Human Eng. + Policy Ind.	44	0.304	0.300	0.297	0.314
5	Human Eng. + LLM Eng.	45	0.239	0.283	0.294	0.249
6	LLM Eng. + Policy Ind.	33	0.309	**0.334**	**0.397**	0.205
7	All Three	61	0.298	0.331	0.356	0.259

Key observations:

- **Policy Induction is the strongest standalone approach** (test F0.5 = 0.290, CV F0.5

= 0.309) with the tightest CV variance (±0.019), though the standalone differences are not statistically significant (see §3).

- **LLM Engineering + Policy Induction is the best overall model** (test F0.5 = 0.334), achieving the highest precision (0.397) of any combination.
- **Human Engineering + LLM Engineering shows no improvement** over either baseline alone (test F0.5 = 0.283), suggesting these feature sets are largely redundant.
- **Adding Human Engineering to LLM Eng. + Policy barely helps** (0.331 vs 0.334), consistent with HQ being partially redundant with both other sets.

---

3. Statistical Significance

We use paired bootstrap testing (10,000 resamples) on the held-out test set to assess whether performance differences are statistically reliable.

Model A	Model B	Delta F0.5	95% CI	p-value
LLM Eng. + Policy	Human Eng.	+0.061	[+0.011, +0.111]	0.007*
All Three	Human Eng.	+0.058	[+0.022, +0.096]	0.001*
LLM Eng. + Policy	LLM Eng.	+0.050	[-0.001, +0.102]	0.028*
All Three	LLM Eng.	+0.047	[+0.001, +0.094]	0.022*
LLM Eng. + Policy	Policy	+0.044	[+0.000, +0.088]	0.025*
Human Eng. + Policy	Human Eng.	+0.028	[-0.006, +0.062]	0.056
Policy	Human Eng.	+0.018	[-0.039, +0.075]	0.278
LLM Eng.	Human Eng.	+0.011	[-0.026, +0.049]	0.273
Policy	LLM Eng.	+0.006	[-0.052, +0.063]	0.422

* *$p < 0.05$*

Key takeaway: No standalone approach significantly outperforms another. However, **combinations involving Policy Induction significantly beat both baselines**: Human Engineering (p = 0.007) and LLM Engineering (p = 0.028). The combination is also significantly better than Policy Induction alone (p = 0.025), confirming genuine synergy rather than one approach doing all the work. See Appendix D for full methodology details including the bootstrap procedure.

---

4. Why Policy Induction Works

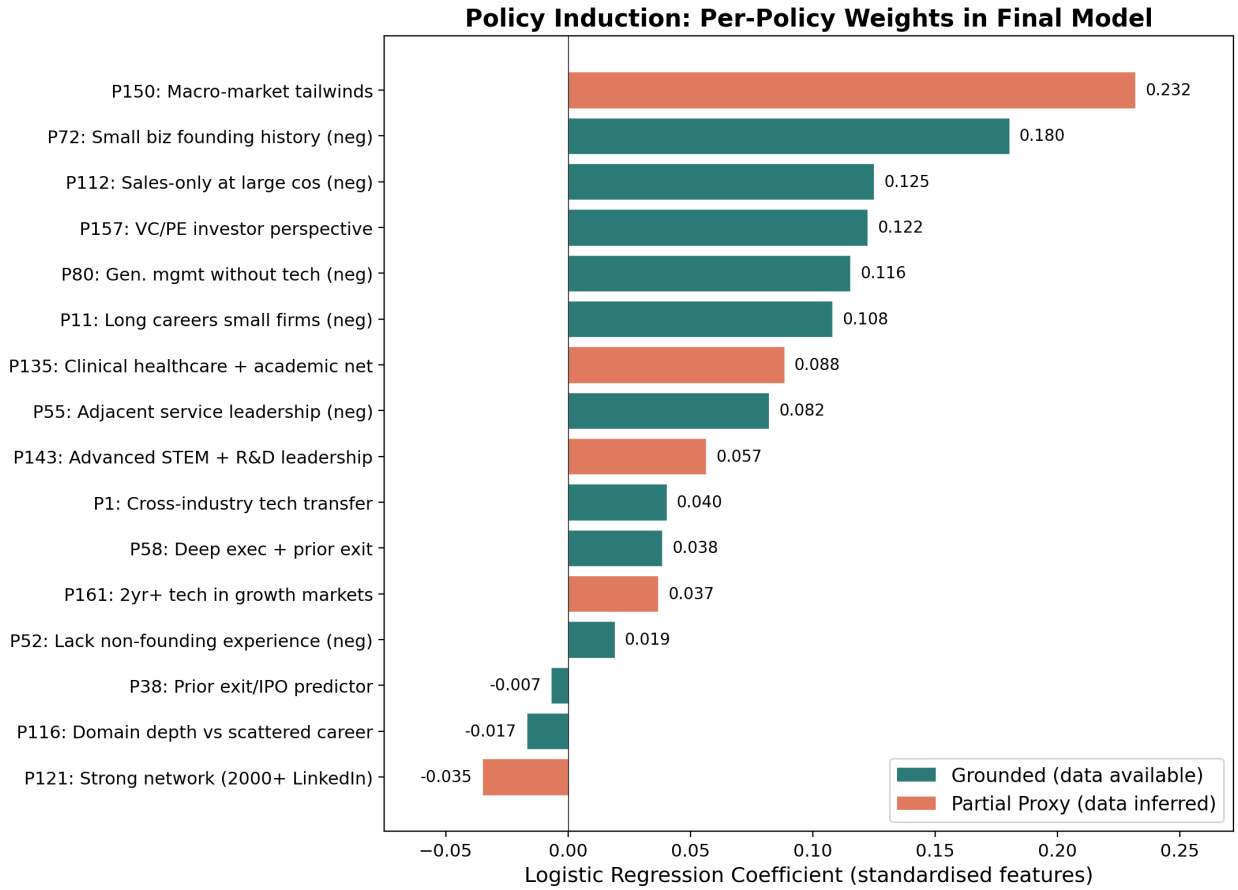

Figure 2: Logistic regression coefficients for the 16-policy model. Teal = grounded policies (reference available data), coral = partial proxy policies (reference unavailable data that the LLM infers from profile signals).

The 16 policies span diverse aspects of founder evaluation: career trajectory, industry expertise, management style, market context, and failure patterns. The L2-regularised logistic regression learns how much to trust each policy's signal.

Top contributors:

- **P150: Macro-market tailwinds** (+0.232). The largest coefficient belongs to a "partial proxy" policy that references macro-market data not present in founder profiles. Despite this, Gemini successfully infers market quality from the founder's industry, timing, and competitive positioning. This is the strongest example of the LLM expanding the effective feature space beyond what is literally available in the data.
- **P72: Small business founding history** (+0.180). A negatively-framed policy: founders with long histories of founding small, non-scaling businesses are less likely to succeed at venture scale. The negative framing creates sharper discrimination than positive-framing policies, because founders who clearly don't match the failure pattern receive high P(success) scores while those who do match receive low scores.
- **P112: Sales-only at large companies** (+0.125). Another negative signal: founders whose

experience is limited to sales roles at large companies lack the technical and operational depth needed for startup success. Interestingly, this policy ranks last (16th) as a standalone predictor but 3rd in the ensemble, because it captures unique signal that no other policy provides. See Appendix A for analysis of why standalone performance can diverge from ensemble importance.

Notable pattern: Negatively-framed policies (P72, P11, P80, P112, P55) dominate the top coefficients. These "red flag" detectors produce sharper score distributions than positive-framing policies, which tend to assign more uniformly optimistic scores. See Appendix C for the full text and diagnostics of all 16 policies.

---

5. Why Combinations Help: The Overlap Story

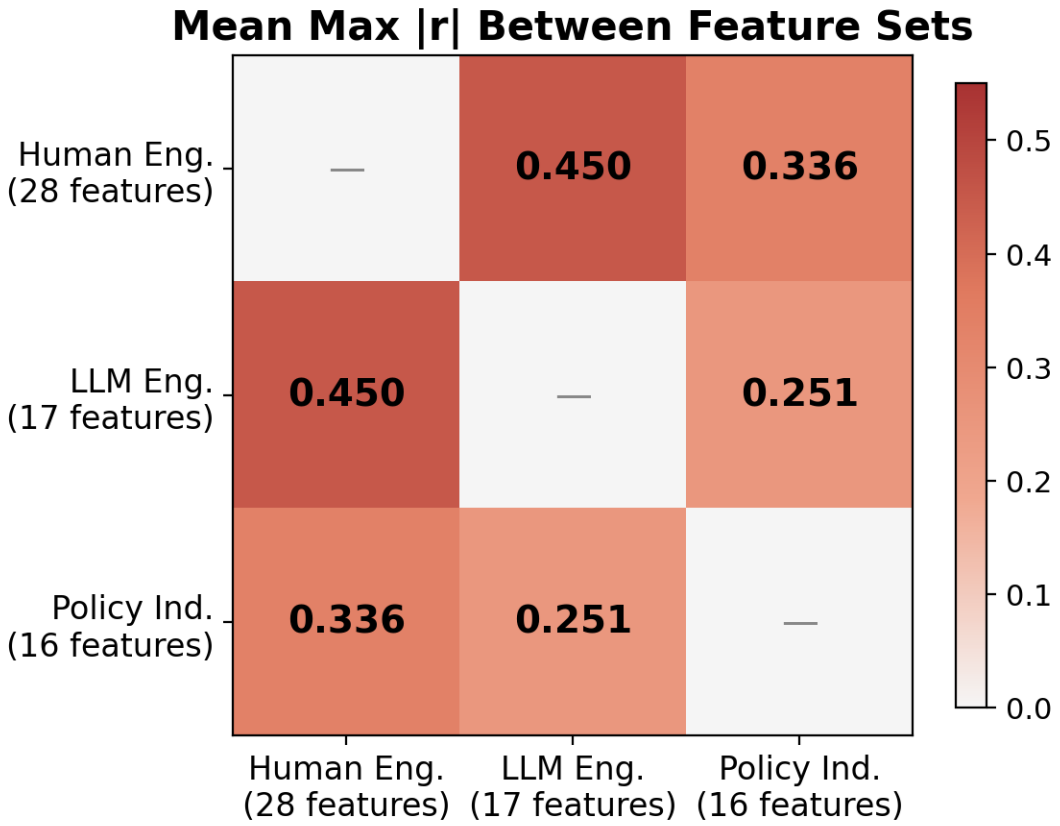

Figure 3: Mean max |Pearson r| between feature sets. Lower values indicate more independent (complementary) feature sets.

The overlap analysis reveals why some combinations work and others don't:

- **Human Engineering and LLM Engineering overlap heavily** (mean max |r| = 0.450). Both derive features from the same structured profile data (education, job tenure, industry), so they capture similar information through different methods. This explains why combining them (Exp 5, F0.5 = 0.283) provides no improvement.
- **Policy Induction has low overlap with LLM Engineering** (mean max |r| = 0.251). 13 of 16 policies have max correlation below 0.3 with any LLM-Engineered feature. This

independence is why LLM Eng. + Policy is the best pairwise combination: each set brings genuinely complementary signal.

- **Policy Induction has moderate overlap with Human Engineering** (mean max |r| = 0.336). Some policies (e.g., P38: prior exit predictor, r = 0.54 with `exit_count`) capture signal already present in HQ features, but most remain distinct.

This overlap pattern is also visible in the results: Human Engineering and LLM Engineering cluster together in both the feature space and in precision/recall performance, while Policy Induction occupies a distinct region (see Figure 4).

---

6. Precision-Recall Tradeoffs

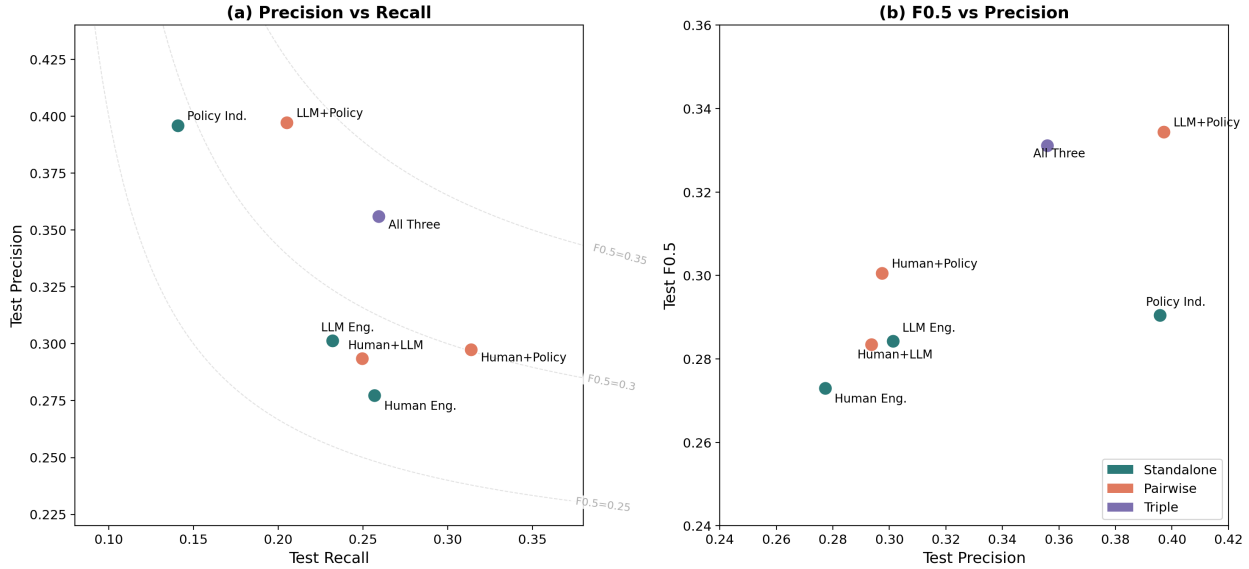

Figure 4: (a) Test precision vs recall with F0.5 iso-lines. (b) Test F0.5 vs precision. Policy-containing approaches occupy the high-precision region.

Panel (a) reveals two distinct strategies:

- **Policy-containing approaches favour precision.** Policy Induction standalone and LLM Eng. + Policy both achieve ~0.40 precision (the highest of any approach) at the cost of lower recall (0.14-0.21). This is desirable for VC screening, where false positives (investing in failures) are more costly than false negatives (missing winners).

- **Human Engineering favours recall.** HQ achieves the most balanced precision/recall tradeoff (0.277/0.257), but at the cost of lower precision. When combined with Policy (Exp 4), recall improves substantially (0.314) but precision drops below either standalone approach.

Panel (b) shows that LLM Eng. + Policy and All Three achieve the best combination of precision and F0.5, occupying the top-right corner and indicating both high precision and high overall performance.

---

7. Grounded vs Partial Proxy Policies

Of the 16 policies, 11 are "grounded" (they reference data directly available in founder profiles, such as job titles, tenure, and exit history) and 5 are "partial proxies" (they reference data not in the profiles, such as LinkedIn connection counts or macro-market conditions).

Both categories perform comparably across all metrics:

Metric	Grounded (n=11)	Partial Proxy (n=5)
Mean pointbiserial r	0.151	0.153
Mean univariate CV F0.5	0.190	0.213
Mean	coefficient	

The top coefficient (P150, +0.232) is a partial proxy; the second (P72, +0.180) is grounded. There is no statistical signature that separates the two groups. This is a positive finding: **the LLM successfully infers proxy signals for data that is not directly available**, expanding the effective feature space beyond what is literally present in the structured profiles.

Appendix A: Univariate Signal vs Model Weight

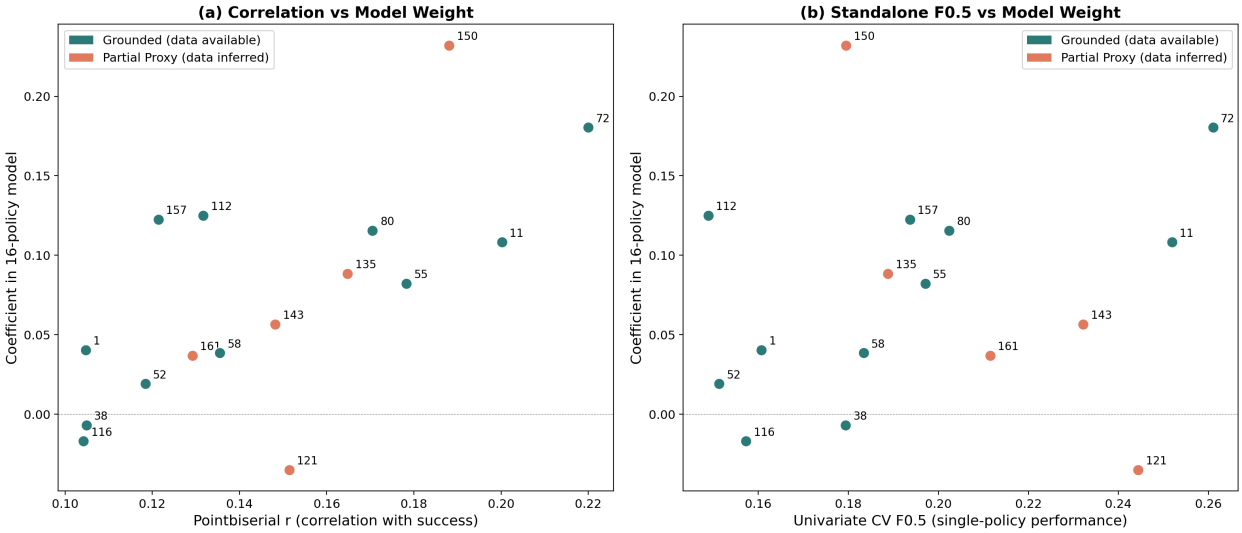

Figure A1: (a) Pointbiserial r vs LR coefficient (strong relationship). (b) Univariate CV F0.5 vs LR coefficient (weak relationship).

Panel (a) shows that pointbiserial r tracks the logistic regression coefficient well (Pearson correlation = 0.68). This is expected: both measure the same underlying quantity (the linear association between a feature and the outcome). The LR coefficient is simply a regularised, multivariate-adjusted version of the correlation.

Panel (b) shows that univariate CV F0.5 is a much weaker predictor of model weight (Pearson correlation = 0.16). Three factors explain this:

1. **F0.5 penalises broad-firing policies.** Policies that flag many founders as likely-to-succeed (high recall) get punished by F0.5's precision weighting, even if their underlying correlation is strong. P112 (recall = 0.75) ranks last by univariate F0.5 but 3rd by coefficient.

2. **Threshold selection noise.** F0.5 requires choosing an optimal decision threshold. For single binary features, this is fragile; some policies have CV variance of ± 0.05, making their F0.5 rank unreliable.

3. **Multivariate vs univariate.** The LR coefficient reflects how much unique information a policy contributes *conditional on all other policies*. A policy with moderate standalone F0.5 but unique signal (e.g., P150) can still receive a large coefficient.

Practical implication: When evaluating individual policy quality, pointbiserial r is a more reliable indicator than standalone F0.5 score. The ensemble model effectively combines diverse weak signals: policies that look mediocre individually can contribute substantial unique signal in combination.

Appendix B: Regularisation Analysis

All logistic regression experiments use L2 regularisation with the regularisation strength C selected via nested cross-validation (inner 3-fold CV) from the grid: C in {0.0005, 0.001, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0, 5.0}.

Why L2?

We evaluated L1, L2, and ElasticNet regularisation:

- **L1** produces sparse models (some coefficients exactly zero) but there is no strong evidence that any policy should be entirely excluded, as all 16 carry some signal.
- **ElasticNet** offers a compromise but adds a second hyperparameter without meaningful improvement.
- **L2** produces more stable coefficients, distributes weight across all features, and with nested C selection avoids overfitting to any single regularisation strength.

The nested CV consistently selects **C = 0.005** for the 16-policy model, indicating relatively strong regularisation is needed.

The "Fewer Is Better" Artifact

Early experiments suggested that smaller subsets of policies outperformed the full set of 16, a "fewer is better" phenomenon. Investigation revealed this was primarily an artifact of two factors:

1. **Regularisation mismatch:** The initial experiments used a fixed C = 0.1, which is too loose for 16 features. At C = 0.01, all 16 policies achieve F0.5 = 0.311, nearly matching the best 3-policy subset (0.313).
2. **Threshold selection instability:** With more features, the optimal classification threshold is harder to estimate from limited CV folds. The 16-policy model loses ~0.020 F0.5 to threshold noise vs ~0.008 for 3-policy models.

With proper nested L2 regularisation, **all 16 policies achieve CV F0.5 = 0.309**, competitive with any subset. The differences between subsets are not statistically significant (bootstrap p = 0.08–0.56). See FEWER_IS_BETTER_REPORT.md for the full analysis.

Appendix C: Individual Policy Descriptions

Each policy is evaluated by Gemini as: *"Given this founder's profile and this policy, predict whether the founder will succeed."* The output P(True) represents P(success | policy, founder), not whether the policy's trait applies. All 16 policies are positively correlated with success.

ID	Description	Category	r(success)	Coeff
150	Macro-market tailwinds favour the startup	Partial Proxy	+0.188	+0.232
72	Long history founding small businesses (neg)	Grounded	+0.220	+0.180
112	Sales-only experience at large companies (neg)	Grounded	+0.132	+0.125
157	VC/PE experience gives investor perspective	Grounded	+0.121	+0.122
80	General management without technical depth (neg)	Grounded	+0.171	+0.116

ID	Description	Category	r(success)	Coeff
11	Long careers in small slow-growth firms (neg)	Grounded	+0.200	+0.108
135	Clinical healthcare + academic-only network	Partial Proxy	+0.165	+0.088
55	Adjacent service leadership, weak for manufacturing (neg)	Grounded	+0.178	+0.082
143	Advanced STEM/business + R&D leadership	Partial Proxy	+0.148	+0.057
1	Cross-industry technical expertise transfer	Grounded	+0.105	+0.040
58	Deep executive experience + prior exit	Grounded	+0.136	+0.038
161	2+ years technical contribution in growth markets	Partial Proxy	+0.129	+0.037
52	Lack of non-founding operational experience (neg)	Grounded	+0.118	+0.019
38	Prior exit/IPO is strongest success predictor	Grounded	+0.105	-0.007
116	Domain depth good; scattered career bad	Grounded	+0.104	-0.017
121	Strong professional network (2000+ LinkedIn)	Partial Proxy	+0.151	-0.035

See POLICY_ABLATION_REPORT.md for detailed diagnostics and conditional success rates.

Full Policy Text

Policy 150 (Partial Proxy, coeff +0.232): > Favor companies operating in macro-markets experiencing significant secular tailwinds, such as resource demand in Extractive Industries, operational efficiency drives, or platform economy growth.

Policy 72 (Grounded, coeff +0.180): > A long history of founding small businesses is a negative predictor for achieving the hyper-growth and massive scale required for top-tier venture outcomes.

Policy 112 (Grounded, coeff +0.125): > Founders with careers limited to sales leadership at large companies often lack the crucial product-building and zero-to-one skills required for early-stage success.

Policy 157 (Grounded, coeff +0.122): > Favor founders whose professional experience includes venture capital or private equity roles, providing an investor-centric perspective and strategic understanding of funding and exit strategies.

Policy 80 (Grounded, coeff +0.116): > General management experience is a negative signal when not paired with deep, recent, and relevant technical expertise required to innovate and scale at venture speed.

Policy 11 (Grounded, coeff +0.108): > Founders with long careers managing small, slow-growth companies often lack the operational experience and mindset required to navigate the distinct challenges of achieving venture-backed hyper-growth and scale.

Policy 135 (Partial Proxy, coeff +0.088): > Founders in the 'Clinical & Diagnostic Healthcare' micro-market with a moderate professional network (e.g., 1000-2000 LinkedIn connections) are high risk if their network is primarily composed of academic peers and research collaborators, lacking robust connections to healthcare industry commercial leaders, experienced healthtech investors, or serial entrepreneurs.

Policy 55 (Grounded, coeff +0.082): > Leadership experience in an adjacent service industry is a weak predictor of success in capital-intensive manufacturing, often indicating gaps in critical operational and go-to-market expertise.

Policy 143 (Partial Proxy, coeff +0.057): > Founders possessing diverse advanced degrees (e.g., MBA combined with PhD/MS in Biological Engineering, Food Science, or Functional Genomics), who demonstrate 6+ years of senior R&D or scientific leadership roles within large enterprises (500+ employees) in Biotechnology, and possessing a robust professional network (2000-5000 LinkedIn connections).

Policy 1 (Grounded, coeff +0.040): > Founders with deep technical expertise from one industry possess a unique advantage when applying it to disrupt a different, more traditional sector.

Policy 58 (Grounded, coeff +0.038): > Prioritize founders with deep executive experience in their target industry, especially when combined with a prior successful company exit, as this de-risks both market and execution challenges.

Policy 161 (Partial Proxy, coeff +0.037): > Founders must have at least 2 years of direct technical contribution (e.g., Software Engineer, Research Scientist) within core technology stacks of high-growth micro-markets (e.g., AI/ML, SaaS Platforms, Cybersecurity), demonstrating critical impact on product development or foundational innovation.

Policy 52 (Grounded, coeff +0.019): > Founders without significant, non-founding operational or industry experience have a lower probability of scaling a company to a large, successful outcome.

Policy 38 (Grounded, coeff -0.007): > A founder's prior success leading a company to a significant exit or IPO is the strongest predictor of their ability to achieve repeatable, large-scale success.

Policy 116 (Grounded, coeff -0.017): > Prioritize founders with deep, sustained expertise in their specific domain. A scattered professional history is a primary indicator of a fatal founder-market fit gap, correlating with failure.

Policy 121 (Partial Proxy, coeff -0.035): > Prioritize founders with a robust professional network (2000+ LinkedIn connections), indicating strong macro-market influence and effective access to capital/talent within high-growth digital micro-segments, especially when combined with deep technical expertise.

Appendix D: Methodology

Nested Cross-Validation

All experiments use 5-fold stratified CV on the training pool. For logistic regression experiments, an inner 3-fold CV selects the regularisation strength C from a grid of 11 values (0.0005 to 5.0). The classification threshold is selected on out-of-fold predictions to maximise F0.5.

Data Pools

- **4,500-founder pool**: Used for experiments involving only Human Engineering and/or Policy Induction features.
- **4,400-founder pool**: Used for experiments involving LLM-Engineered features. 100 founders are held out as seed examples for the LLM feature generation pipeline.
- **Test set**: All experiments produce predictions on the same 4,500-founder held-out test set, enabling direct comparison.

Rule Override

Experiments including Human Engineering features apply a deterministic rule: founders with `exit_count > 0` are given a score of 1.0 (always predicted positive). This reflects strong domain knowledge that prior exits are the single strongest success signal.

Bootstrap Testing

10,000 paired bootstrap resamples of the test set. For each resample, F0.5 is computed for both models, and the difference is recorded. The one-sided p-value is the fraction of resamples where the observed positive difference reverses (i.e., the better model appears worse). Significance threshold: $p < 0.05$.

Standardisation

Continuous features are standardised to zero mean and unit variance before logistic regression. Binary features are left unstandardised. XGBoost (used only for Human Engineering) does not require standardisation.